NSCLC Clinical Trials
Investment and Risk Assessment Supporting Exhibits

Source data: AACT Clinical Trials Database
Analysis: Excel

Trial Volume and Enrollment by Phase

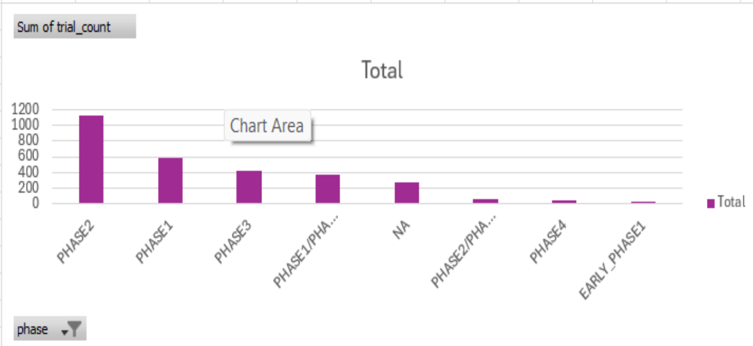
Trial counts and average enrollment size for each NSCLC trial phase, pulled from the AACT dataset (Query results, raw table).

phase	trial_count	avg_enrollment
PHASE2	1130	83.35377778
PHASE1	590	82.74872666
	518	1201.264228
PHASE3	418	454.4628297
PHASE1/PHASE2	376	117
NA	273	180.0586081
PHASE2/PHASE3	58	513.7068966
PHASE4	50	411.4285714
EARLY_PHASE1	33	27.60606061

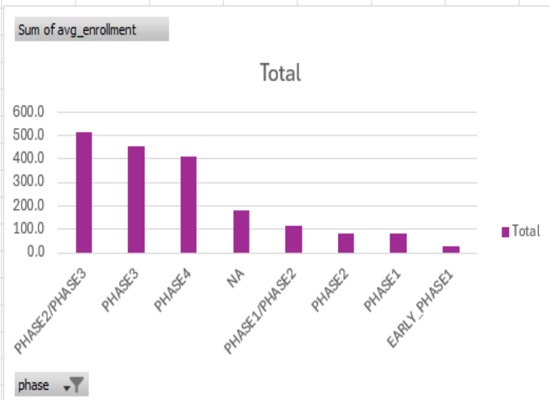
Trial Count and Enrollment by Phase Charts

Pivot charts visualizing trial count and average enrollment across phases. Phase 2 dominates trial volume; Phase 2/3 and Phase 3 trials enroll the largest patient populations on average.

Row Labels	Sum of trial_count
PHASE2	1130
PHASE1	590
PHASE3	418
PHASE1/PHASE2	376
NA	273
PHASE2/PHASE3	58
PHASE4	50
EARLY_PHASE1	33
Grand Total	**2928**



Row Labels	Sum of avg_enrollment
PHASE2/PHASE3	513.7
PHASE3	454.5
PHASE4	411.4
NA	180.1
PHASE1/PHASE2	117.0
PHASE2	83.4
PHASE1	82.7
EARLY_PHASE1	27.6
Grand Total	**1870.4**



Termination Rate by Phase

Total trials, terminated trials, and termination rate (%) by phase. Termination risk falls sharply from Phase 1 (21.5%) to Phase 4 (2%), consistent with standard drug-development risk curves.

phase	total_trials	terminated_trials	termination_rate_pct
PHASE1	590	127	21.5
EARLY_PHASE1	33	6	18.2
PHASE1/PHASE2	376	63	16.8
PHASE2/PHASE3	58	7	12.1
PHASE2	1130	130	11.5
NA	273	22	8.1
PHASE3	418	34	8.1
PHASE4	50	1	2
	518	0	0

Trial Count by Sponsor and Phase

Pivot table breaking down trial count by sponsor organization across each phase. Identifies which companies (e.g., AstraZeneca, Hoffmann-La Roche) have full-pipeline presence across all phases.

Sum of trial_count	Column Labels								
Row Labels	EARLY_PHASE1	NA	PHASE1	PHASE1/PHASE2	PHASE2	PHASE2/PHASE3	PHASE3	PHASE4	Grand Total
National Cancer Institute (NCI)		3	26	11	44	1	8		93
Hoffmann-La Roche			7	5	35	2	34	7	90
AstraZeneca		1	12	4	19		22	4	62
Eli Lilly and Company			12	7	29		12	1	61
Sun Yat-sen University		2	1	3	38	2	9	1	56
Merck Sharp & Dohme LLC			7		8	1	28		44
Bristol-Myers Squibb			4	6	15	2	14	1	42
Novartis Pharmaceuticals			16	4	10		8		38
Betta Pharmaceuticals Co., Ltd.			8	4	12	1	6	6	37
Pfizer		1	12	4	10		7	1	35
Genentech, Inc.			9	1	16		3	1	30
Memorial Sloan Kettering Cancer Center		2	6	4	14				26
Amgen			5	3	10		7	1	26
Tianjin Medical University Cancer Institute and Hospital		1	3		18	2	1		25
M.D. Anderson Cancer Center		2	5	6	11				24
Hunan Province Tumor Hospital				1	15	1	3		20
Washington University School of Medicine	1	2	5	3	8				19
AbbVie			9	1	2	3	3		18
Daiichi Sankyo			3	3	8		4		18
Shanghai Chest Hospital		3		3	6	3	2	1	18
Sichuan Baili Pharmaceutical Co., Ltd.			2	3	5	2	5		17
Massachusetts General Hospital	1	7	2	2	5				17
Samsung Medical Center		1			11		3	1	16
Shanghai Pulmonary Hospital, Shanghai, China					14	1		1	16
Chia Tai Tianqing Pharmaceutical Group Co., Ltd.			3	2	4	2	4		15
Grand Total	**2**	**25**	**157**	**80**	**367**	**23**	**183**	**26**	**863**

TAM / SAM / SOM Market Sizing Tagrisso (AstraZeneca)

Market sizing model for Tagrisso in EGFR-mutation-positive NSCLC. Walks from total addressable lung cancer cases down to AstraZeneca's serviceable obtainable market, in patients and revenue.

TAM/SAM/SOM Model	
Tagrisso (AstraZeneca), NSCLC	Tagrisso specifically targets EGFR-mutation-positive NSCLC
US New Lung Cancer Cases (2026)	229410
NSCLC Share of Lung Cancers	85%
EGFR Mutation Prevalence (US NSCLC)	15%
Treatment Rate Assumption	70%
AstraZeneca Market Share Assumption	65%
Tagrisso Annual Cost	$279,420.00

In ref to SOM
Treatment rate: not every eligible patient gets diagnosed/treated (testing gaps, access issues) a reasonable assumption is ~70%
AstraZeneca's market share among treated EGFR+ patients since Tagrisso is the standard first-line therapy in this space, a reasonable assumption is ~60-65% share (they're the clear leader, but not the only

MARKET SIZING - PATIENTS	
TAM (NSCLC patients)	194998.5
SAM (EGFR+ patients)	29249.775
SOM (AstraZeneca captured patients)	13308.64763

MARKET SIZING - REVENUE	
TAM ($)	$54,486,480,870.00
SAM ($)	$8,172,972,130.50
SOM ($)	$3,718,702,319.38

Tier	Value
TAM	$54,486,480,870.00
SAM	$8,172,972,130.50
SOM	$3,718,702,319.38

TAM (Total Adderasable Market)	224,390 total lung cancer × 85% NSCLC = ~190,730 NSCLC patients/yr
SAM (Serviceable Addressable Market)	190,730 × 15% EGFR-mutation rate = ~28,610 eligible patients/yr
SOM (Serviceable Obtainable Market)	28,610 × 70% treated × 65% market share ≈ ~13,000 patients/yr

Probability of Success (PoS) Model NSCLC

Compares industry-benchmark oncology approval likelihood against observed AACT termination rates by phase, and derives a cumulative probability-of-success estimate with stated assumptions and caveats.

Probability of Success Model NSCLC

Industry Benchmark (Oncology)

Oncology Phase 1 → Approval (Likelihood of Approval)	5%
Non-Oncology Phase 1 → Approval (for comparison)	12%

Observed Termination Rates (AACT NSCLC Dataset, Query 4)

Early Phase 1	18.80%
Phase 1	21.50%
Phase 1/2	17%
Phase 2	11.50%
Phase 2/3	12.10%
Phase 3	8.10%
Phase 4	2%

Derived Cumulative PoS (Using Observed Data)

Implied Phase 1 Success Rate (1 - Termination)	78.50%
Implied Phase 3 Success Rate (1 - Termination)	92%
Simple Cumulative (Phase 1 × Phase 3 survival, illustrative only)	72.141500%

Metric	Value
Oncology LOA (Industry)	5%
Non-Oncology LOA (Industry)	12%
Derived Cumulative	72%

Interpretation:

- Industry data shows oncology drugs have a substantially lower likelihood of approval from Phase 1 (~5%) than non-oncology therapeutic areas (~12%)
- Observed AACT termination rates are used here as a rough proxy for phase survival, not a true transition-probability calculation
- This NSCLC dataset shows directionally consistent risk with the broader oncology pattern
- The true likelihood of approval (5%, industry-sourced) is meaningfully lower than the simple derived cumulative (~72%), because it accounts for additional failure modes beyond termination — e.g., trials that complete but still fail to demonstrate efficacy.